Sales Forecasting & Demand Intelligence System

Executive Business Report | Prepared by: Tanushree Parakh

Executive Summary

This report summarizes a new system that uses four years of our sales history to predict future demand, flag unusual sales spikes or drops, and group products by how their demand behaves. The system helps the business estimate future product demand, identify unusual sales patterns, and support better inventory planning. This reduces the chances of stock shortages, excess inventory, and improves decision-making through an easy-to-use dashboard.

Key Findings from the Data

Looking back over four years of sales history, three patterns stood out:

- **Revenue by category —** Technology products generate the highest revenue overall, with Furniture and Office Supplies also contributing meaningfully to total sales.
- **Clear seasonality —** Sales follow a clear seasonal pattern, indicating that customer demand remains consistent during certain months of the year and can be planned for effectively.
- **Best-performing model selected —** We tested three forecasting approaches (SARIMA, Prophet, and XGBoost) and measured their accuracy using standard error metrics. The most accurate model was selected to generate the forecasts below.

3-Month Sales Forecast

The selected model projects stable sales over the next three months, with no major spike or drop expected. Each forecast is accompanied by a confidence range that represents the expected variation around the predicted sales value. These confidence ranges help determine appropriate inventory levels and improve inventory planning.

Top 3 Anomalies Detected

Using two independent detection methods (Isolation Forest and Z-Score analysis) to cross-check each other, three notable anomalies were flagged in the historical data:

1. **Demand spike — likely promotional.** A sharp, short-term spike in sales lined up with a promotional or festive period — expected and explainable, but worth planning inventory buffers around in future years.
2. **Demand drop — likely supply disruption.** A sudden drop in sales that most likely reflects a stockout or supply chain disruption rather than an actual drop in customer demand.
3. **Unusual spike week — likely bulk order or discount.** One week showed unusually high sales, consistent with a bulk order or a special discount event rather than typical organic demand.

Recommendation: Monitor these types of anomalies regularly so that unexpected business events can be identified and addressed quickly.

Product Demand Segmentation & Stocking Strategy

Products were grouped into four segments based on sales volume, growth rate, volatility, and average order value:

Segment	What it means	Recommended stocking strategy
High Volume, Stable Demand	Consistently strong sellers with predictable demand	Keep sufficient stock on hand with regular, scheduled replenishment
Growing Demand	Sales trending upward month over month	Increase stock gradually to stay ahead of the trend
Low Volume, High Volatility	Unpredictable, infrequent demand	Hold limited inventory; monitor closely and reorder reactively
Declining Demand	Sales trending downward	Reduce inventory and use promotions to clear remaining stock

Business Recommendations

1. **Adopt forecast-based inventory planning —** Set reorder points directly from the model's forecast and confidence range instead of relying only on historical averages. This is backed by the forecasting results above and directly reduces both stockouts and excess inventory.
2. **Align stock levels to demand segment —** Use the growth-rate and volume data from the segmentation analysis to guide purchasing — build up stock ahead of demand for the Growing segment, and phase down orders for the Declining segment before it ties up working capital.
3. **Institutionalize anomaly monitoring —** Build a weekly review of the anomaly detection output into the supply chain team's routine, so unexpected shifts (like the three flagged above) are caught and acted on within days, not months.

Risk & Limitation

This system learns from historical sales data, so it cannot fully capture unexpected events such as an economic downturn, a natural disaster, a major unplanned promotion, or a sudden shift in the market. Forecasts should be treated as a strong planning baseline, not a guarantee, and the model should be retrained regularly as new sales data comes in to keep its accuracy from drifting over time.

Supporting Dashboard

Alongside the analysis, an interactive dashboard was built so managers can track actual sales against the forecast, review flagged anomalies, and see which segment each product currently falls into — all without needing to open any code. This gives both supply chain and finance teams a shared, real-time view of demand for day-to-day decision-making, rather than relying on a static monthly report.

Conclusion & Next Steps

Overall, the Sales Forecasting & Demand Intelligence System provides a clear, data-driven view of future demand, helping the business improve inventory planning, respond quickly to unexpected sales patterns, and make informed supply chain decisions. To put these findings into practice, we recommend the following next steps:

1. **Pilot the forecast in inventory planning —** Roll out forecast-based reorder points for the highest-revenue category (Technology) first, then extend to Furniture and Office Supplies.
2. **Formalize anomaly review ownership —** Assign ownership of the weekly anomaly report to the supply chain team, with a clear escalation path to procurement when a disruption-type anomaly appears.
3. **Schedule regular model refreshes —** Retrain the forecasting model on a quarterly basis, or sooner if a major market shift occurs, to keep predictions accurate as conditions change.

With these steps in place, the system moves from a one-time analysis to an ongoing decision-support tool the business can rely on for planning inventory, managing cash tied up in stock, and responding quickly when demand doesn't follow the plan.